

Learning Market Signals from Financial News Headlines Using Interpretable Machine Learning

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WEBER STATE
UNIVERSITY

About ME (Mathematician by love, statistician by training and do data science for \$\$!)



Talk outline

0

~35 min



Presentation

Questions?

Your questions/comments are welcome at any time.



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Talk outline

1. Research Motivation and Questions
2. Data and ML Pipeline Overview
3. Exploratory Text Analysis
4. Topic Modeling (LDA, BERTopic)
5. Signal Engineering and Feature Construction
6. Classification Results and Dataset Complexity
7. Model-Agnostic Interpretability (MDI, Permutation, SHAP)
8. Conclusions and Future Directions

Motivation & Research Questions

Motivation

- ❑ Financial markets are highly sensitive to new information
- ❑ Headlines encode structured signals about macroeconomic risk, monetary policy, and geopolitical uncertainty
- ❑ Existing machine learning (ML) models often lack interpretability

Research Question

- ❑ Can we extract economically meaningful, interpretable signals from news headlines to predict market direction?

Data & ML Pipeline Overview

THE WALL STREET JOURNAL.

World Business U.S. Politics Economy Tech Markets & Finance Opinion Free Expression Arts

2026 > January > January 1

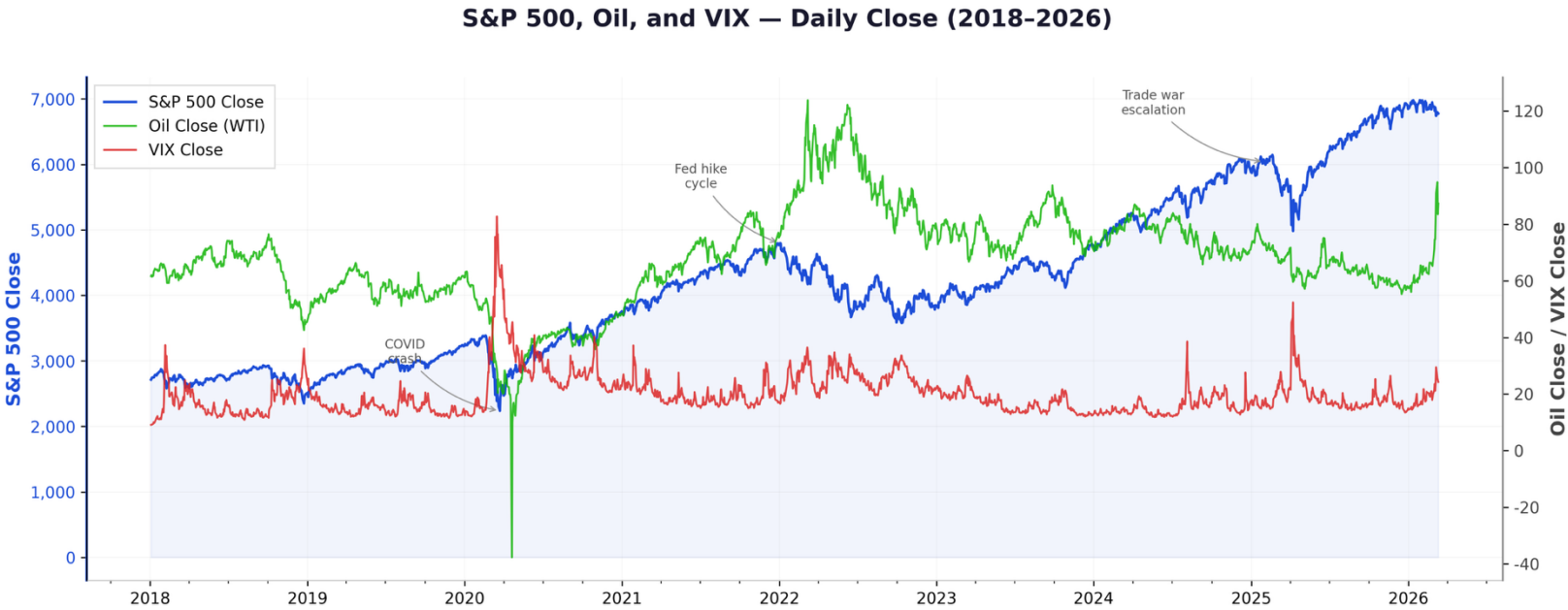
Archive for January 1, 2026

Golf, Peace Negotiations and Parties Fill Trump's Mar-a-Lago Holiday

Baidu's AI Chip Unit Kunlunxin Plans Hong Kong Listing

Modi Taps India's Consumer Power in Trade Fight With Trump

Indiana Turns the Tide on Alabama in Rose Bowl Blowout



Data & ML Pipeline Overview

WSJ Headline Corpus

- ❑ Jan 2018 – Mar 2026
- ❑ 2,992 trading days
- ❑ 8+ years of daily coverage

DOCUMENTS PROCESSED

292k

News headlines

TOKEN VOLUME

2.6M

Total words in corpus

VOCABULARY SIZE

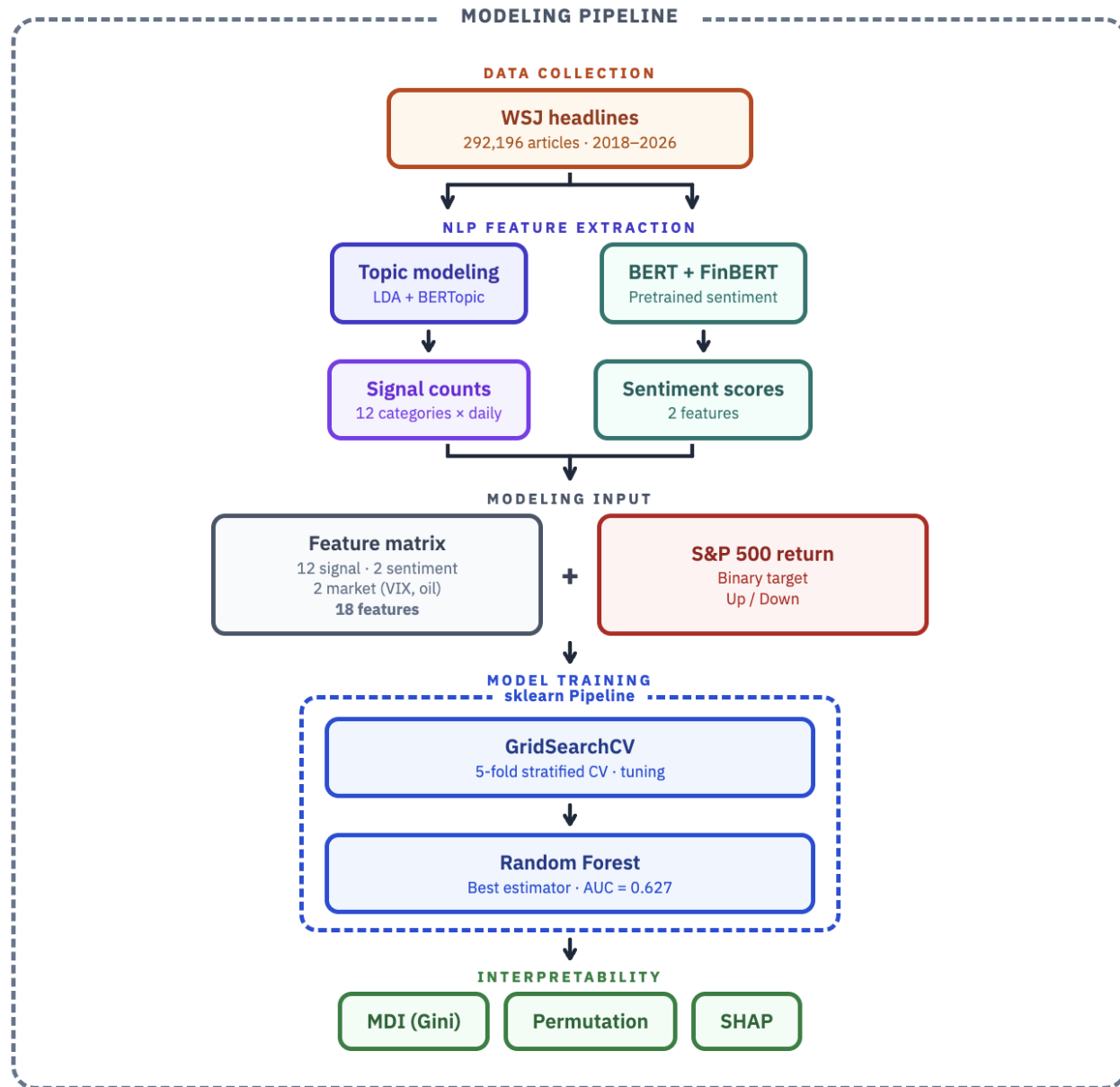
80k

Unique terms

AVG. DAILY VOLUME

~98

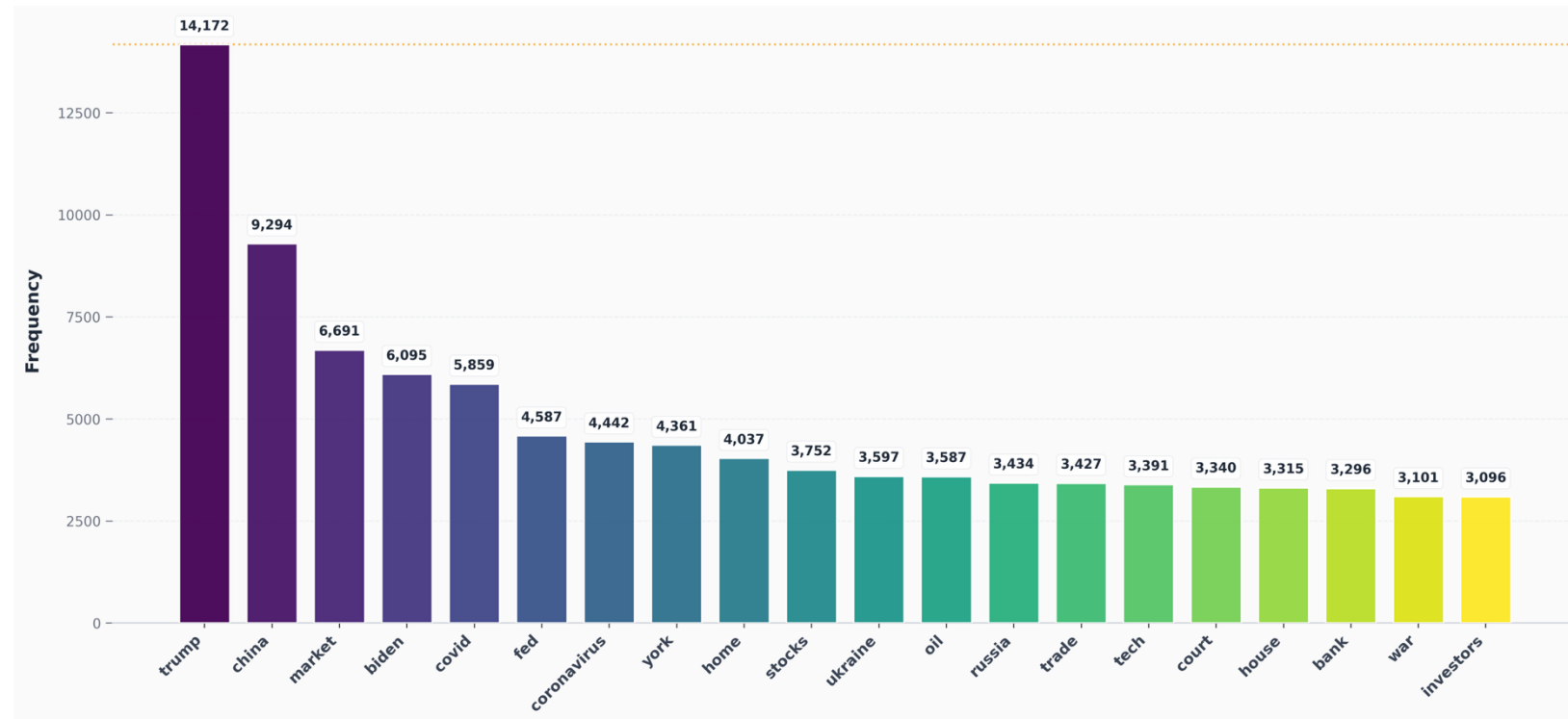
Headlines per trading day



Dominant Terms in Financial News

Exploratory Text Analysis

- ❑ Headlines are driven by politics, policy, and global events (Trump, China, Fed)
- ❑ COVID and geopolitical conflict (Ukraine, war) are highly prevalent
- ❑ Frequent terms like market, stocks, investors reflect continuous financial interpretation of news



Topic Structure in Financial Headlines

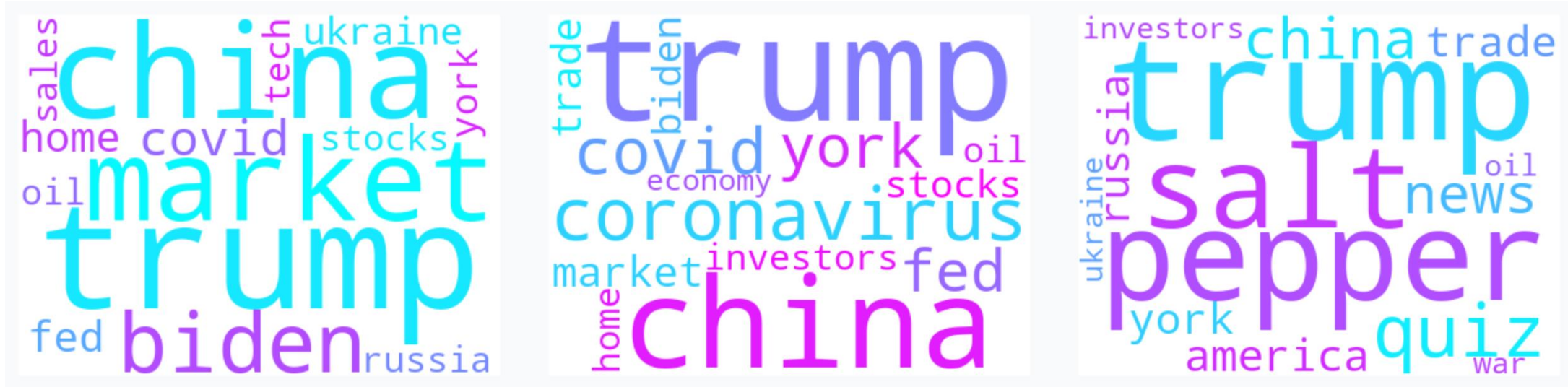
- ❑ Latent Dirichlet Allocation (LDA): Unsupervised generative model that represents each headline as a mixture of topics and each topic as a distribution over words



- ❑ Four dominant macroeconomic narratives drive investor behavior: Fed/trade policy, geopolitical conflict, pandemic & domestic politics, and commodity/tariff markets
- ❑ These latent topics directly informed the construction of keyword-based signal categories for the feature matrix

Semantic Clustering of Headlines

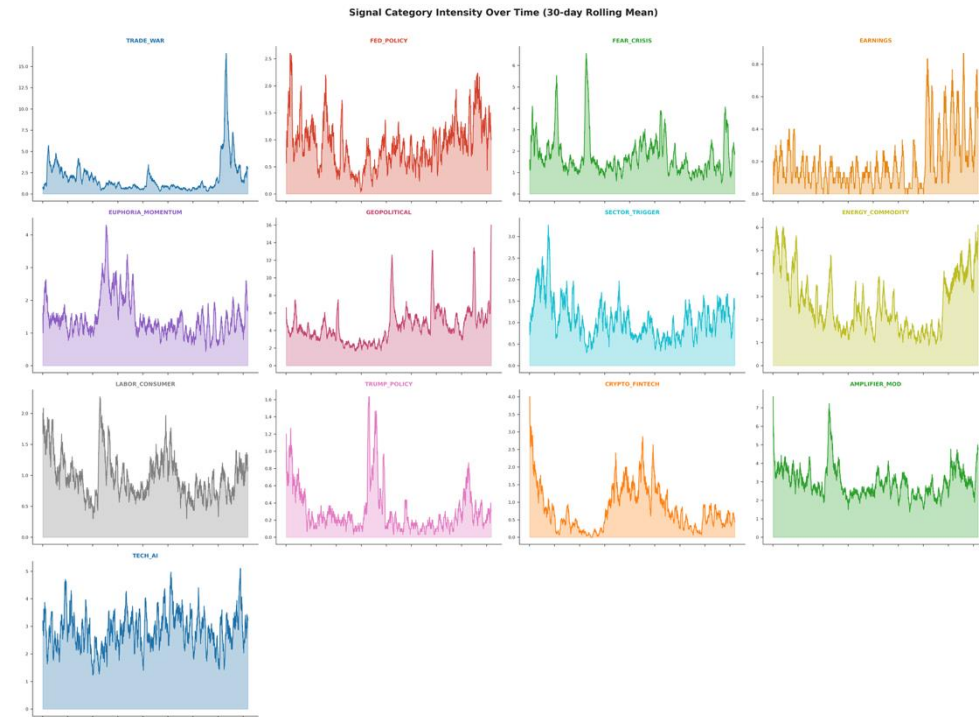
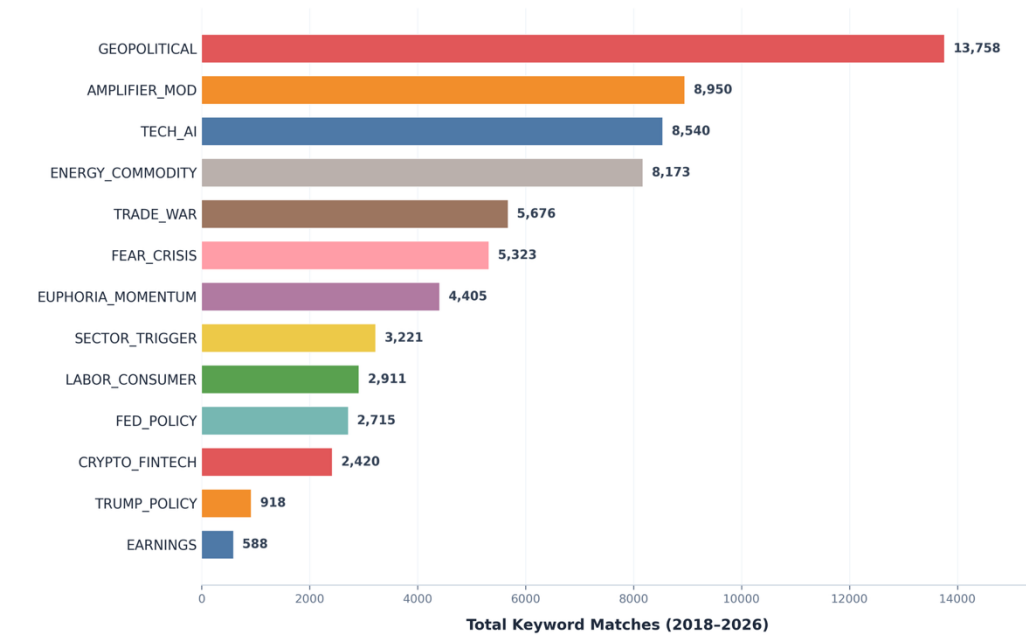
- ❑ BERTopic (BERT-based): Leverages BERT contextual embeddings to cluster headlines by semantic similarity



- ❑ Three clusters align with LDA themes — political economy and crisis/policy response with an additional noise cluster capturing non-financial lifestyle content (e.g., food, entertainment) from broader WSJ coverage

Feature Engineering: From Topic Words to Feature Matrix

- ❑ 12 keyword dictionaries derived from LDA/BERTopic and domain knowledge
- ❑ Daily keyword frequencies form a sparse signal vector capturing market attention by macro theme

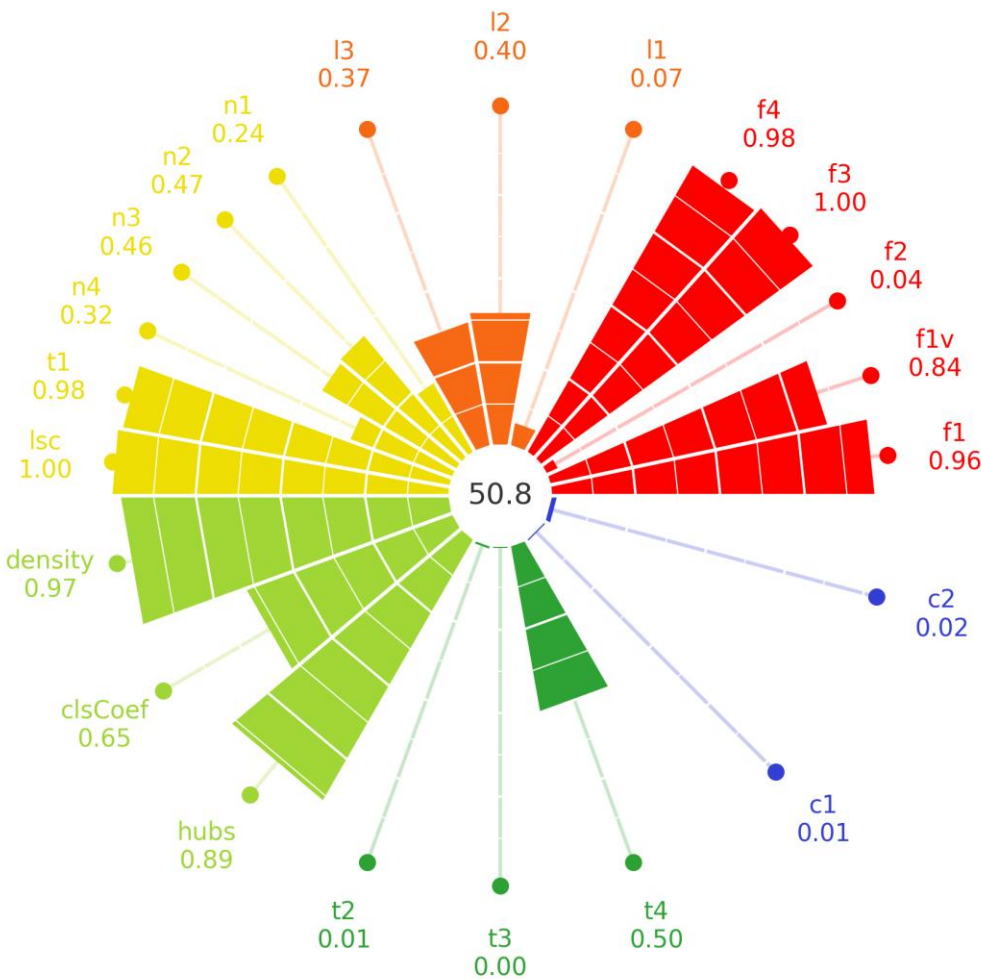
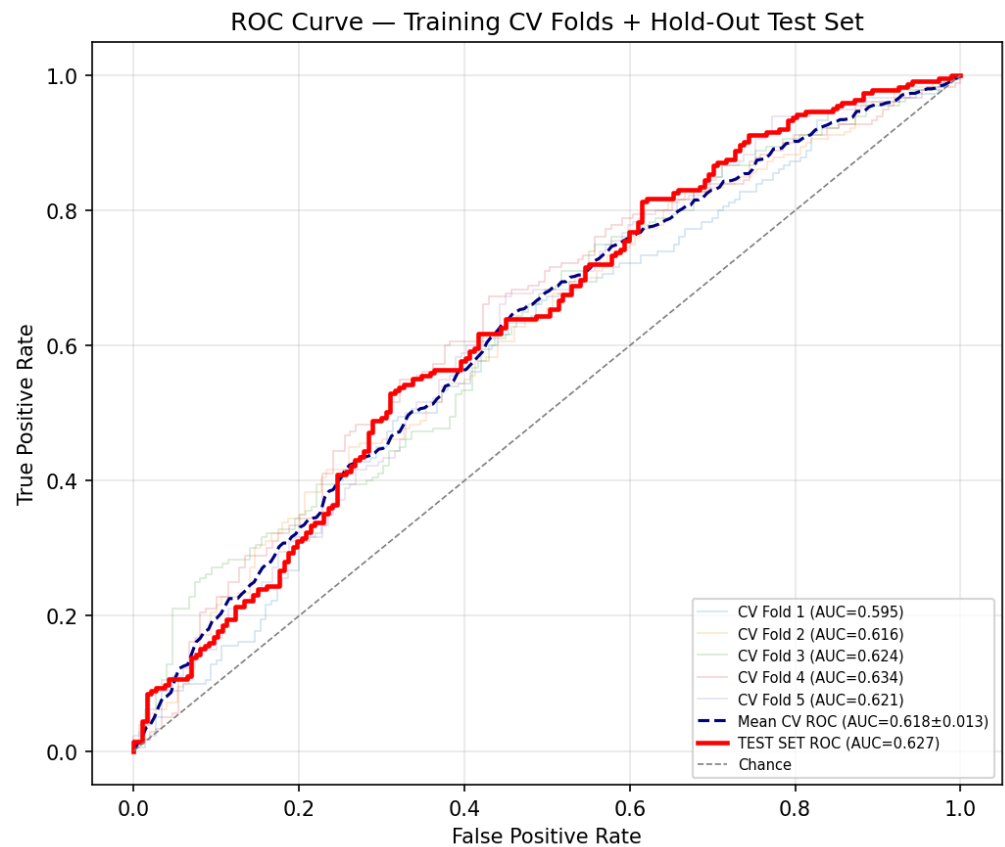


- ❑ 16 features: 12 headline signals, sentiment scores (BERT, FinBERT), and market controls (oil, VIX)



Results: Random Forest Model

☐ Stable out-of-sample performance, no overfitting

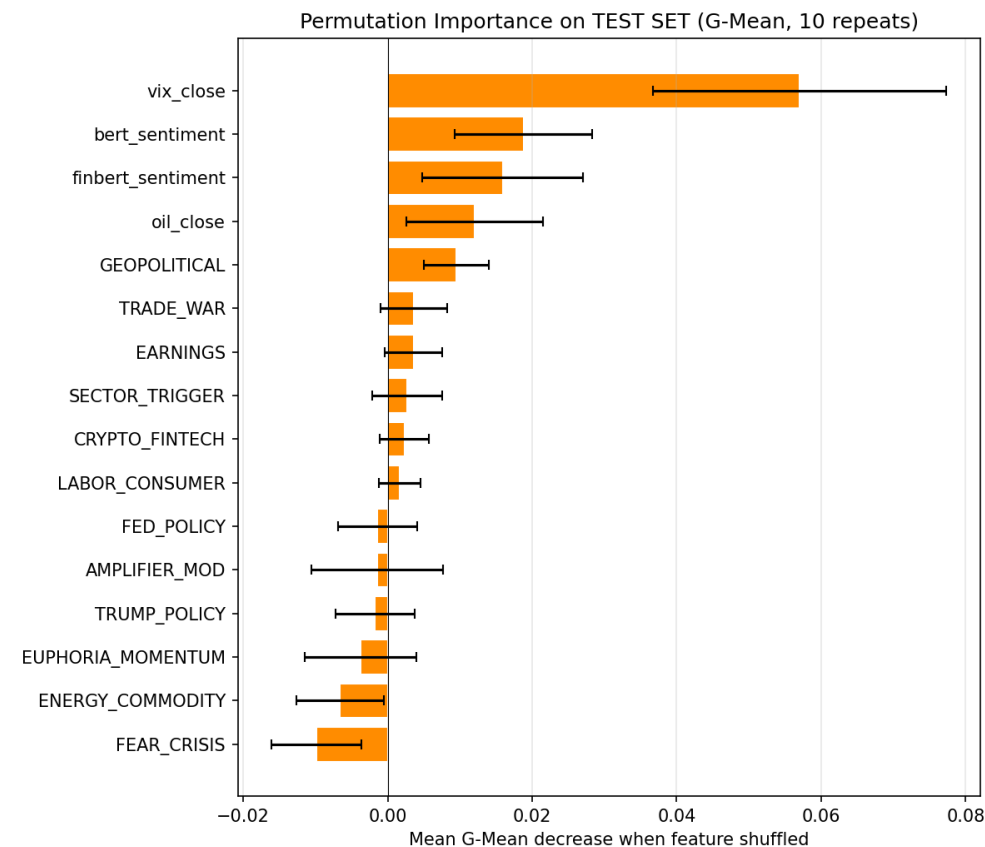
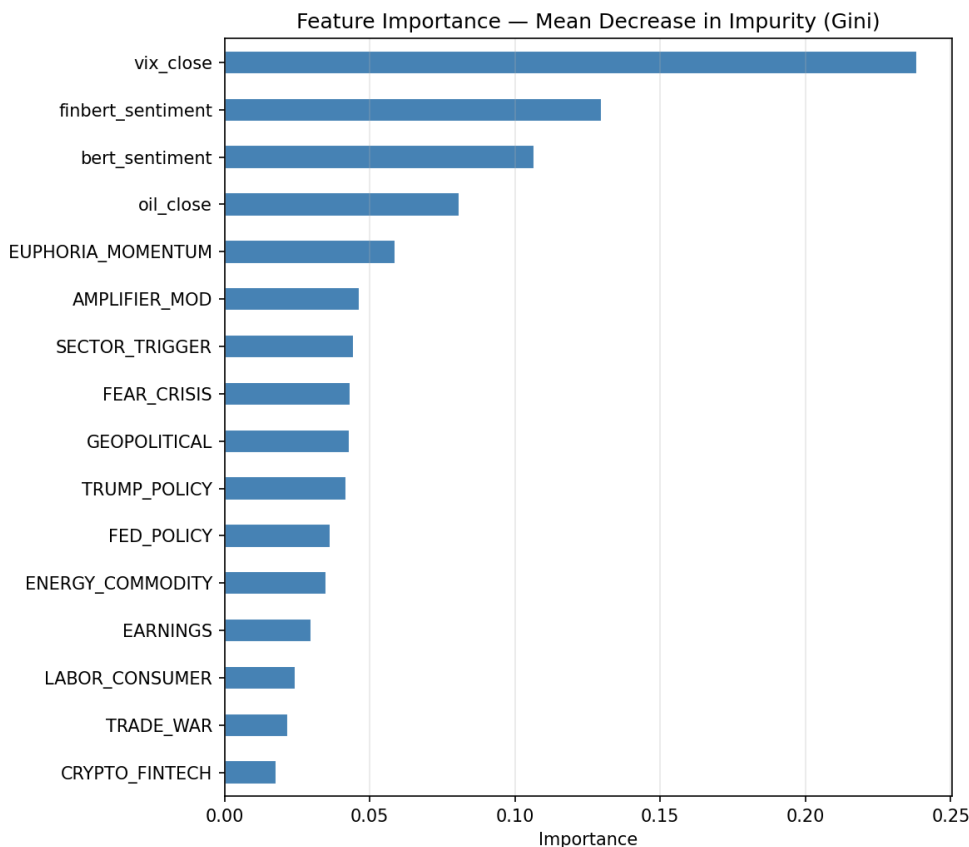


☐ Complexity analysis confirms daily market direction is inherently difficult to classify



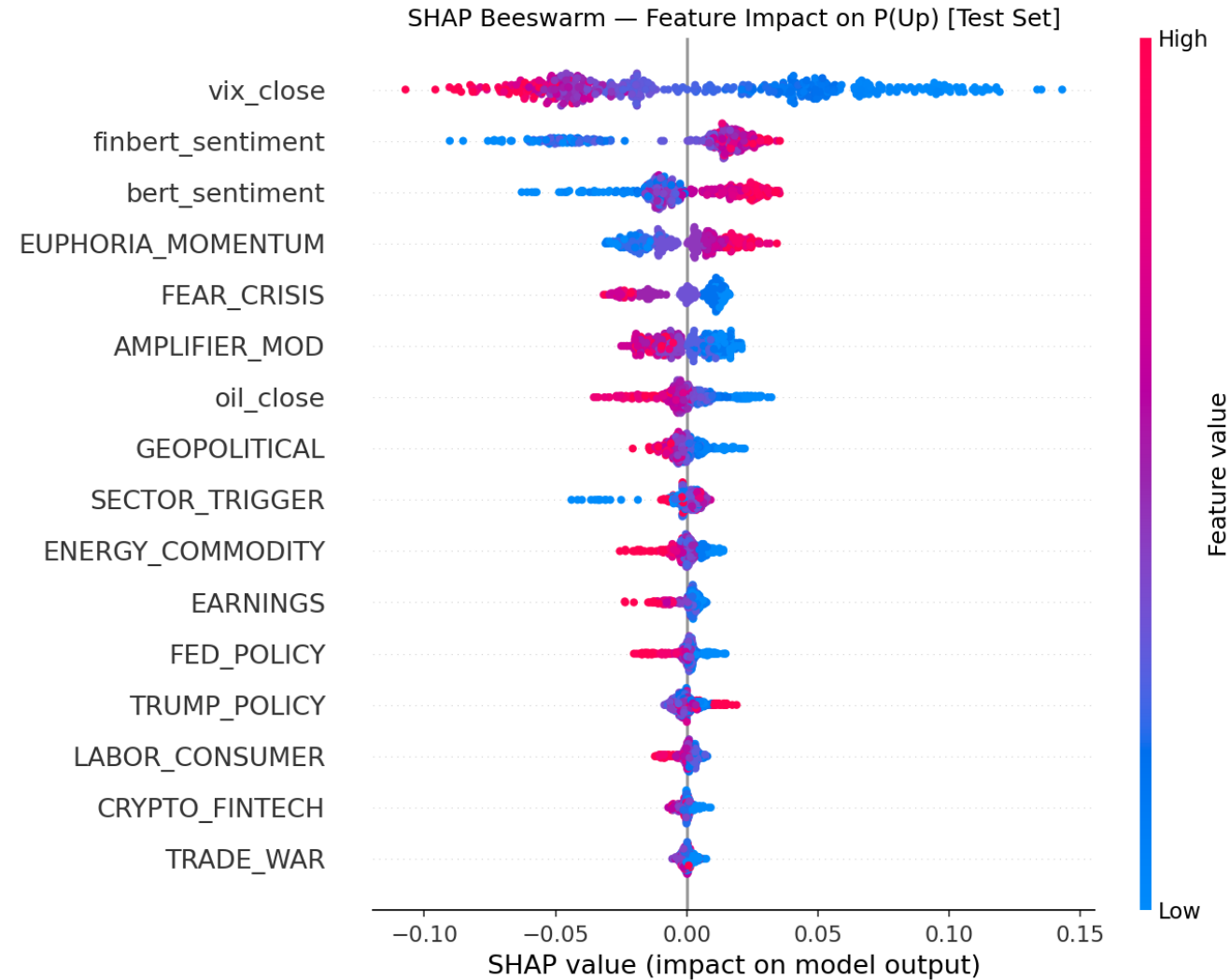
Feature Importance: MDI vs Permutation

- ❑ Market-implied volatility (VIX) dominates next-day S&P 500 direction; sentiment (BERT, FinBERT) and oil act as secondary factors
- ❑ Headline-based signals provide limited incremental explanatory power



Feature Importance: SHAP

- ❑ Market direction is primarily driven by volatility (VIX) and sentiment, while headline-based signals offer limited incremental explanatory power
- ❑ Higher FinBERT and BERT sentiment scores push predictions toward Up, confirming bullish headline tone carries measurable predictive signal for market direction



Conclusions and Future Directions

Conclusions

- ❑ LDA and BERTopic extract interpretable macroeconomic themes from 292K WSJ headlines, forming economically meaningful signal categories for direction prediction
- ❑ Market direction is driven primarily by implied volatility (VIX), commodity prices (oil), and transformer-based sentiment — consistent across MDI, permutation, and SHAP
- ❑ SHAP provides transparent, instance-level attribution of how each feature influences classification decisions

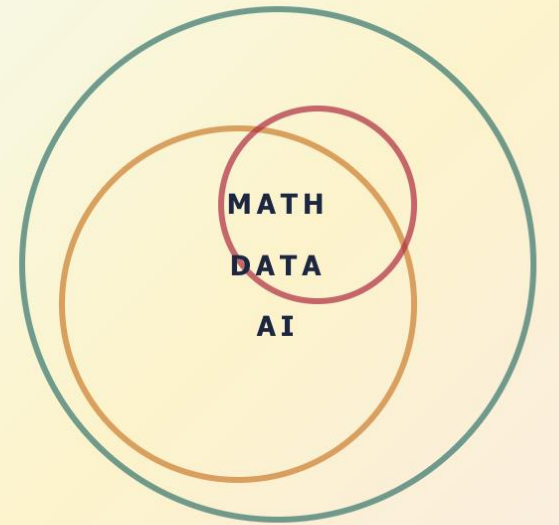
Future Directions

- ❑ Develop an agentic AI framework integrating financial statements, earnings calls, and news headlines for multi-source equity prediction

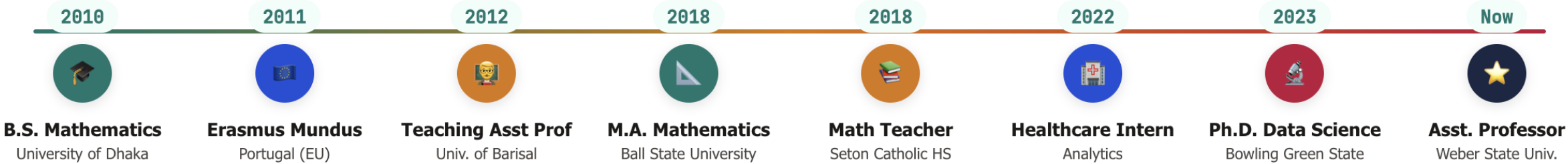


My Academic & Professional Journey

From Mathematics to Data Science — Academia, Industry, and AI



My Journey at a Glance



Dr. Jishan Ahmed '23 is the first to earn a doctorate in data science from BGSU.

BGSU awards first doctorate in data science four years after launching rigorous program to meet workforce demands

Estimated Reading Time: 3 minutes

[BGSU](#) / [News](#) / [2023](#) / [October](#) / BGSU awards first doctorate in data science four years after launching rigorous program to meet workforce demands

Succeeding in Job Interviews

1 Invest in Teaching Training

- Teaching seminars at Ball State built real classroom confidence
- Helped me secure my first teaching job in the U.S.
- Practical skills outweigh theoretical credentials

2 Why I Started Getting Interviews

- Shifted from theory → real-world data problems
- Focused on communicating impact, not just models
- Told a compelling story about my unique path



How I Got My First Teaching Job (U.S.)



Step 01

Where I Applied

- Handshake
- School career websites
- Private school recruiters



Step 02

My Strategy

- Private & charter schools (no license required)
- Tailored each application



Step 03

Application Materials

- CV + Cover Letter
- Teaching Statement (critical)
- Strong references



Step 04

What Helped Me

- Teaching training at Ball State
- Feedback from Career Center
- Peer review of materials



Teaching Job Search: Schools & Colleges



Where to Apply

- Handshake
- School/district websites
- Private school recruiters
- College career pages

 carneysandoe.com

 glassdoor.com

 indeed.com

 ivytech.edu



Key Differences

- Private/charter → often no license required
- Public schools → may hire provisionally
- Some sponsor H1B / green card (case-by-case)



Required Materials

- Teaching statement / philosophy
- CV + cover letter
- Strong references
- Tailored for each application



Data Science & Industry Job Search



Where to Apply

● LinkedIn (primary)

● Indeed

● Glassdoor

● Company career pages



What Matters Most

- Projects (GitHub portfolio)
- Internships & hands-on experience
- Referrals & networking
- Demonstrated impact over degree name



The Power of Networking



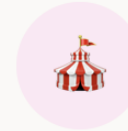
References Open Doors

Strong references move your resume to the top of the stack



Start with Your Circle

Professors, peers, alumni become mentors and collaborators



Go to Conferences

SciPy, IEEE, and academic & industry events build visibility



Follow Up

A short personalized message builds long-term connections



Skills That Get You Hired

⚡ Technical

- Python
- R
- SQL
- Git / GitHub
- Cloud (AWS)

📊 Analytical

- Model Understanding
- Domain Knowledge
- Tableau
- Power BI

💼 Professional

- Communication
- Teamwork
- Continuous Learning
- Teaching & Mentoring

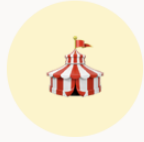


Build Experience — Start Now



Work with Professors

Seek research opportunities and collaborate on publications



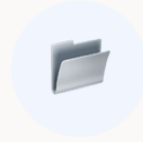
Attend Conferences

Present your work, learn, and network with peers



Take Internships

Any experience helps — industry exposure is invaluable



GitHub Portfolio

Build a strong portfolio showcasing real projects

Smart Job Search Playbook

01

Quality Over Quantity

20 strong applications beat 100 generic ones

02

Build References Early

Cultivate strong relationships before you need them

03

Network for Visibility

Use connections to get in front of hiring managers

04

Practice Explaining Your Work

Clear communication sets you apart from others

05

Ask for Help from Mentors

Don't be afraid to reach out — mentors want to help you succeed. Their guidance can transform your search.



What I'd Do Differently



Network Earlier

Start building professional connections from day one of your program



Learn Industry Tools

Pick up SQL, Tableau, and cloud platforms sooner rather than later



Pursue Internships

Seek hands-on industry experience as early as possible



Online Presence

Build your GitHub, LinkedIn, and blog to showcase your work



What Helped Me Stand Out



Openness to New Technologies

Embracing change and learning new tools with enthusiasm



Theory + Real-World

Combining mathematical foundations with practical applications



Data-Driven Teaching

Using data to improve teaching methods and student outcomes



Continuous Learning

Never stopping — always growing skills and knowledge



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Who Shaped My Career

🎓 *"My Ball State professors influenced me the most. Their mentorship, guidance, and belief in my potential transformed my academic and professional trajectory."*





Thank You

Questions?



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